

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)
2025-26, Term-III
Project Problem Statement Submission Form

Sec No- 1	Team No-	Project Title- Airline-Crew-Scheduling-UI
Team Lead Roll Number		2520080015
Team Lead Name		Veridhi Hari Priya
Team Member Roll Number		2520030399
Team Member Name		Kuchipudi Harini

1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

The Airline Crew Scheduling UI project focuses on developing a responsive and user-friendly web interface for managing airline crew operations. Crew scheduling is a complex task that involves organizing pilot and cabin crew assignments, tracking duty hours, handling leave requests, and ensuring compliance with aviation regulations. Traditional systems can be difficult to manage and lack intuitive visualization, making the process time-consuming.

This project addresses these challenges by providing a frontend-based dashboard that simplifies crew management through an interactive and visually structured interface. The system includes features such as roster views, duty tracking, leave request forms, and real-time alerts for scheduling conflicts. Users can easily navigate through different sections, update schedules, and monitor crew activities efficiently.

The application is built using modern web technologies such as HTML5, CSS3, JavaScript, and React.js to ensure responsiveness and smooth user interaction. It uses component-based design and state management to simulate dynamic behavior and real-time updates without relying on a backend system.

Overall, the project aims to enhance usability and efficiency by transforming complex scheduling tasks into an accessible and interactive experience. It demonstrates how a well-designed frontend interface can significantly improve workflow management in airline crew operations.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
UI Design & Layout (HTML, CSS, Structure, Styling)	2520080015	
Frontend Functionality (JavaScript, Interactivity, UI Logic)	2520030399	

3. Module Details

(Explain the responsibility of each student concerning their assigned module.)

Module 1: UI Design & Layout (HTML, CSS, Styling)

Student with Roll Number **2520080015** is responsible for designing the overall user interface of the Airline Crew Scheduling dashboard. This includes creating page structures using HTML and styling them with CSS to ensure a responsive and visually appealing design. The student develops components such as navigation bars, dashboards, forms, and scheduling views while maintaining consistency and usability.

Module 2: Frontend Functionality (JavaScript, React.js, UI Logic)

Student with Roll Number **2520030399** is responsible for implementing the interactive features of the application using JavaScript and React.js. This includes handling user interactions, managing component state, implementing features like roster updates, duty tracking, and notifications, and ensuring smooth navigation using React Router. The student also uses LocalStorage and mock data to simulate real-time functionality.

4. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

- HTML5 – For structuring the application
- CSS3 – For styling and responsive design
- JavaScript (ES6+) – For interactivity and dynamic behavior
- React.js – For building a component-based user interface
- Axios – For handling API requests (mock or future integration)
- React Router – For navigation between pages
- Context API / Redux – For state management
- LocalStorage – For storing user/session data
- Bootstrap / Tailwind CSS – For responsive UI design
- npm – For package management
- Visual Studio Code – Development environment
- Web Browser (Chrome/Edge) – For testing and execution

Guide Name and Signature (remarks if any) _____